Naive Bayes Classifier

Application of Bayes Rule

AIDS test (Bayes rule)

Data

- **Approximately 0.1% are infected**
- **Test detects all infections**
- **Test reports positive for 1% healthy people**

Probability of having AIDS if test is positive

$$P(a=1|t=1) = \frac{P(t=1|a=1)P(a=1)}{P(t=1)}$$

$$= \frac{P(t=1|a=1)P(a=1)}{P(t=1|a=1)P(a=1) + P(t=1|a=0)P(a=0)}$$

$$= \frac{1 \cdot 0.001}{1 \cdot 0.001 + 0.01 \cdot 0.999} = 0.091$$

Only 9%!...

slide by Barnabás Póczos & Aarti

Improving the diagnosis

Use a weaker follow-up test!

- Approximately 0.1% are infected
- Test 2 reports positive for 90% infections
- Test 2 reports positive for 5% healthy people

$$P(a=0|t_1=1,t_2=1) = \frac{P(t_1=1,t_2=1|a=0)P(a=0)}{P(t_1=1,t_2=1|a=1)P(a=1)+P(t_1=1,t_2=1|a=0)P(a=0)}$$

$$= \frac{0.01 \cdot 0.05 \cdot 0.999}{1 \cdot 0.9 \cdot 0.001 + 0.01 \cdot 0.05 \cdot 0.999} = 0.357$$

$$P(a=1|t_1=1,t_2=1) = 0.643$$

64%!...

AIDS test (Bayes rule)

Why can't we use Test 1 twice?

- Outcomes are **not** independent,
- but tests 1 and 2 **conditionally independent (by assumption)**:

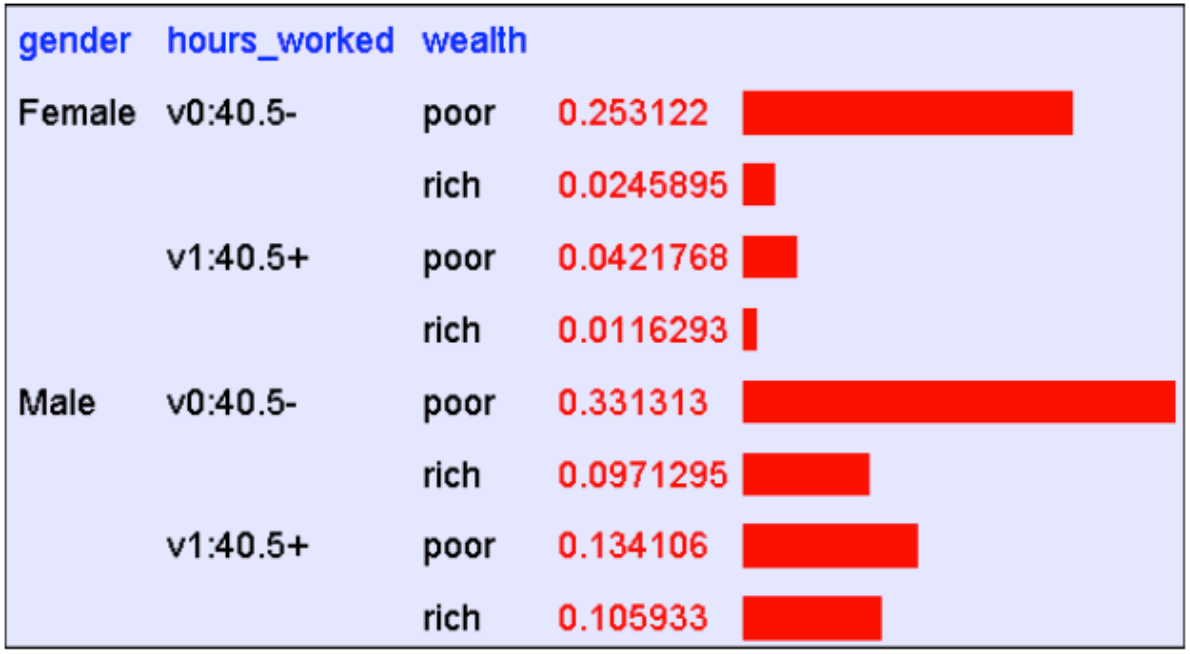
$$p(t_1, t_2|a) = p(t_1|a) \cdot p(t_2|a)$$

The Naïve Bayes Classifier

Let's learn classifiers by learning P(Y|X)

Consider Y=Wealth, X=<Gender, HoursWorked>

gender	hours_worked	wealth	
Female	v0:40.5-	poor	0.253122
		rich	0.0245895
	v1:40.5+	poor	0.0421768
		rich	0.0116293
Male	v0:40.5-	poor	0.331313
		rich	0.0971295
	v1:40.5+	poor	0.134106
		rich	0.105933



We want to estimate the probability that a person is rich or poor given the gender and the number of hours worked

Gender	HrsWorked	P(rich | G,HW)	P(poor | G,HW)
F	<40.5	.09	.91
F	>40.5	.21	.79
M	<40.5	.23	.77
M	>40.5	.38	.62

The first number in the Table is P(F, 40-, poor)

P(poor | F, 40-) = P(F, 40-,poor) / P(F,40-) = (0.253122 / (0.253122+0.0245895) =0.91

How many parameters must we estimate?

Suppose $X = \langle X_1, \ldots X_n \rangle$
where X_i and Y are boolean RV's

Gender	HrsWorked	P(rich | G,HW)	P(poor | G,HW)
F	<40.5	.09	.91
F	>40.5	.21	.79
M	<40.5	.23	.77
M	>40.5	.38	.62

To estimate $P(Y| X_1, X_2, \ldots X_n)$

$$2^n$$

If we have 30 boolean X_i's: $P(Y | X_1, X_2, \ldots X_{30})$

$$2^{30}$$

Naïve Bayes Assumption

Naïve Bayes assumption: Features X_1 and X_2 are conditionally independent given the class label Y:

$$P(X_1, X_2|Y) = P(X_1|Y)P(X_2|Y)$$

More generally:

$$P(X_1 \ldots X_d|Y) = \prod_{i=1}^{d} P(X_i|Y)$$

Naïve Bayes Assumption, Example

Task: Predict whether or not a picnic spot is enjoyable

Training Data: $X = (X_1 \quad X_2 \quad X_3 \quad \ldots \quad \ldots \quad X_d) \quad Y$

n rows

Sky	Temp	Humid	Wind	Water	Forecst	EnjoySpt
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

Naïve Bayes Assumption, Example

Task: Predict whether or not a picnic spot is enjoyable

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Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

Naïve Bayes assumption:

$$P(X_1 \ldots X_d | Y) = \prod_{i=1}^{d} P(X_i | Y)$$

How many parameters to estimate?

(X is composed of d binary features,
Y has K possible class labels)

$(2^d\text{-}1)K$ vs $(2\text{-}1)dK$

Naïve Bayes Classifier

Given:

- Class prior $P(Y)$
- d conditionally independent features $X_1,\ldots X_d$ given the class label Y
- For each X_i feature, we have the conditional likelihood $P(X_i|Y)$

Naïve Bayes Decision rule:

$$f_{NB}(\mathbf{x}) = \arg\max_y P(x_1,\ldots,x_d \mid y)P(y)$$

$$= \arg\max_y \prod_{i=1}^{d} P(x_i|y)P(y)$$

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Naïve Bayes in a Nutshell

Bayes rule:

$$P(Y = y_k | X_1 \dots X_n) = \frac{P(Y = y_k) P(X_1 \dots X_n | Y = y_k)}{\sum_j P(Y = y_j) P(X_1 \dots X_n | Y = y_j)}$$

Assuming conditional independence among X_i's:

$$P(Y = y_k | X_1 \dots X_n) = \frac{P(Y = y_k) \prod_i P(X_i | Y = y_k)}{\sum_j P(Y = y_j) \prod_i P(X_i | Y = y_j)}$$

So, to pick most probable Y for $X^{new} = \Re X_1, \dots, X_n \top$

$$Y^{new} \leftarrow \arg\max_{y_k} P(Y = y_k) \prod_i P(X_i^{new} | Y = y_k)$$

Naïve Bayes Algorithm for discrete features

$$f_{NB}(\mathbf{x}) = \arg\max_y \prod_{i=1}^{d} P(x_i|y)P(y)$$

We need to estimate these probabilities!

Estimators

For Class Prior $$\widehat{P}(y) = \frac{\{\#j : Y^{(j)} = y\}}{n}$$

For Likelihood $$\frac{\widehat{P}(x_i, y)}{\widehat{P}(y)} = \frac{\{\#j : X_i^{(j)} = x_i, Y^{(j)} = y\}/n}{\{\#j : Y^{(j)} = y\}/n}$$

NB Prediction for test data: $X = (x_1, \ldots, x_d)$

$$Y = \arg\max_y \widehat{P}(y) \prod_{i=1}^{d} \frac{\widehat{P}(x_i, y)}{\widehat{P}(y)}$$

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Subtlety: Insufficient training data

What if you never see a training instance where $X_1 = a$ when $Y = b$?

For example,

there is no X_1='Earn' when Y='SpamEmail' in our dataset.

$$\Rightarrow P(X_1 = a, Y = b) = 0 \Rightarrow P(X_1 = a | Y = b) = 0$$

$$\Rightarrow P(X_1 = a, X_2 ... X_n | Y) = P(X_1 = a | Y) \prod_{i=2}^{d} P(X_i | Y) = 0$$

Thus, no matter what the values $X_2, \ldots, X_d$ take:

$$P(Y = b \mid X_1 = a, X_2, \ldots, X_d) = 0$$

What now???

Naïve Bayes Alg — Discrete features

Training data: $\{(X^{(j)}, Y^{(j)})\}_{j=1}^{n}$ $\quad X^{(j)} = (X_1^{(j)}, \ldots, X_d^{(j)})$

Use your expert knowledge & apply prior distributions:

- Add m "virtual" examples
- Same as assuming conjugate priors

Assume priors: $Q(Y = b)$ $\quad Q(X_i = a, Y = b)$

MAP Estimate:

$$\widehat{P}(X_i = a | Y = b) = \frac{\{\#j : X_i^{(j)} = a, Y^{(j)} = b\} + mQ(X_i = a, Y = b)}{\{\#j : Y^{(j)} = b\} + mQ(Y = b)}$$

virtual examples with Y = b

called Laplace smoothing

Estimating Parameters: Y, X_i discrete-valued

Maximum likelihood estimates:

$$\hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\}}{|D|}$$

$$\hat{\theta}_{ijk} = \hat{P}(X_i = x_j | Y = y_k) = \frac{\#D\{X_i = x_j \wedge Y = y_k\}}{\#D\{Y = y_k\}}$$

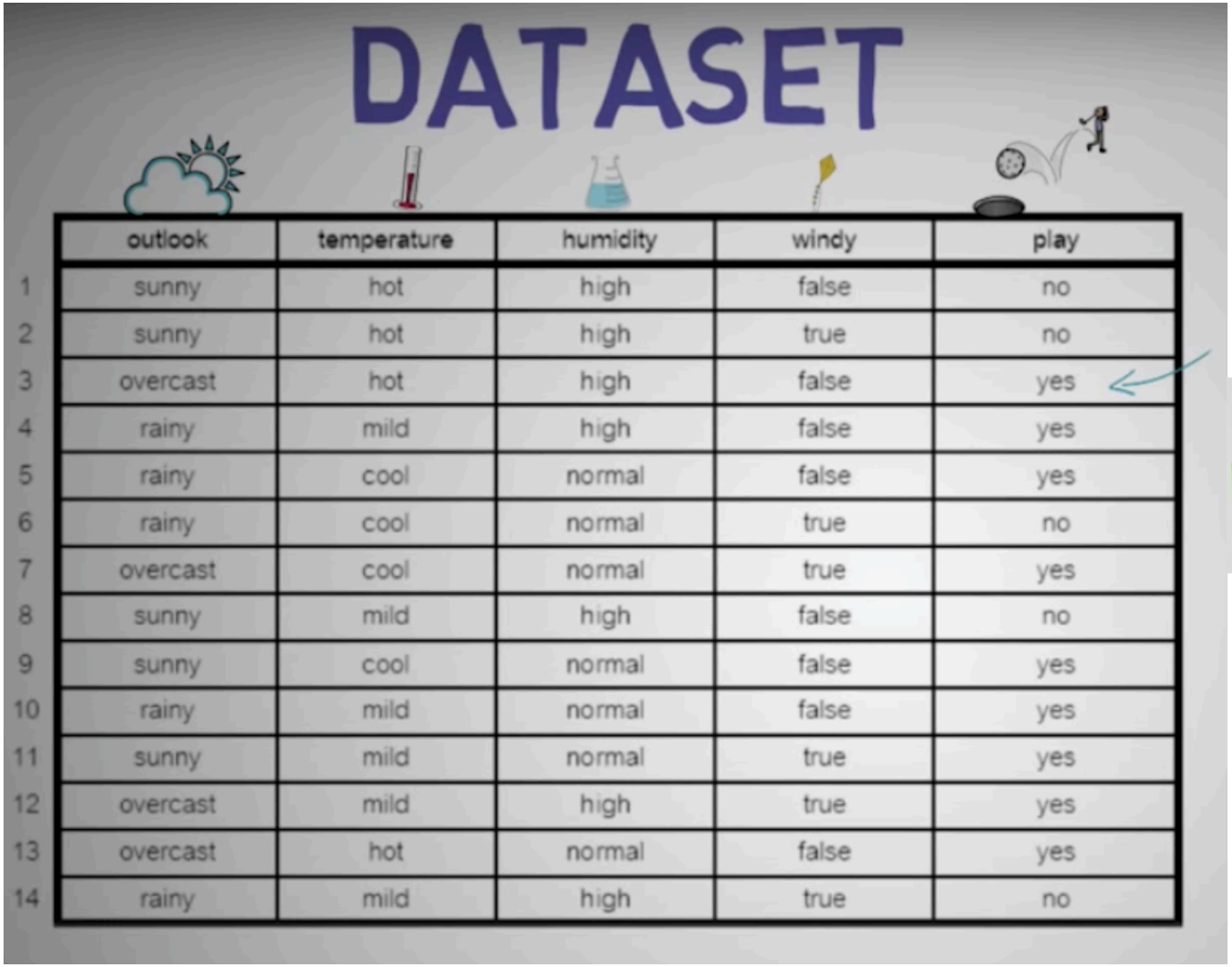
MAP estimates (Beta, Dirichlet priors):

$$\hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\} + (\beta_k - 1)}{|D| + \sum_m (\beta_m - 1)}$$

$$\hat{\theta}_{ijk} = \hat{P}(X_i = x_j | Y = y_k) = \frac{\#D\{X_i = x_j \wedge Y = y_k\} + (\beta_k - 1)}{\#D\{Y = y_k\} + \sum_m (\beta_m - 1)}$$

Only difference: "imaginary" examples

Naive Bayes Example



	outlook	temperature	humidity	windy	play
1	sunny	hot	high	false	no
2	sunny	hot	high	true	no
3	overcast	hot	high	false	yes
4	rainy	mild	high	false	yes
5	rainy	cool	normal	false	yes
6	rainy	cool	normal	true	no
7	overcast	cool	normal	true	yes
8	sunny	mild	high	false	no
9	sunny	cool	normal	false	yes
10	rainy	mild	normal	false	yes
11	sunny	mild	normal	true	yes
12	overcast	mild	high	true	yes
13	overcast	hot	normal	false	yes
14	rainy	mild	high	true	no

Goal: predict the probability of playing (yes/no) given the weather conditions.

Frequency Tables

1.

Outlook	Yes	No	P(Yes)	P(No)
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rainy	3	2	3/9	2/5
Total	**9**	**5**	**100%**	**100%**

2.

Temperature	Yes	No	P(Yes)	P(No)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
Total	**9**	**5**	**100%**	**100%**

3.

Humidity	Yes	No	P(Yes)	P(No)
High	3	4	3/9	4/5
Normal	6	1	6/9	1/5
Total	**9**	**5**	**100%**	**100%**

4.

Wind	Yes	No	P(Yes)	P(No)
FALSE	6	2	6/9	2/5
TRUE	3	3	3/9	3/5
Total	**9**	**5**	**100%**	**100%**

5.

Play		P(Yes) / P(No)
Yes	9	9/14
No	5	5/14
Total	**14**	**100%**

The first number is the joint probability P(Sunny, Yes). P(Sunny | Yes) = 2/9

Naive Bayes Example

- Given
  - Outlook: Sunny
  - Temperature: Cool
  - Humidity: High
  - Wind: Strong
  - Shall we play or not?

Probability that we can play the game.
> P(Outlook=Sunny | Play=Yes) = 2/9
> P(Temperature=Cool | Play=Yes) = 3/9
> P(Humidity=High | Play=Yes) = 3/9
> P(Wind=Strong | Play=Yes) = 3/9
> P(Play=Yes) = 9/14

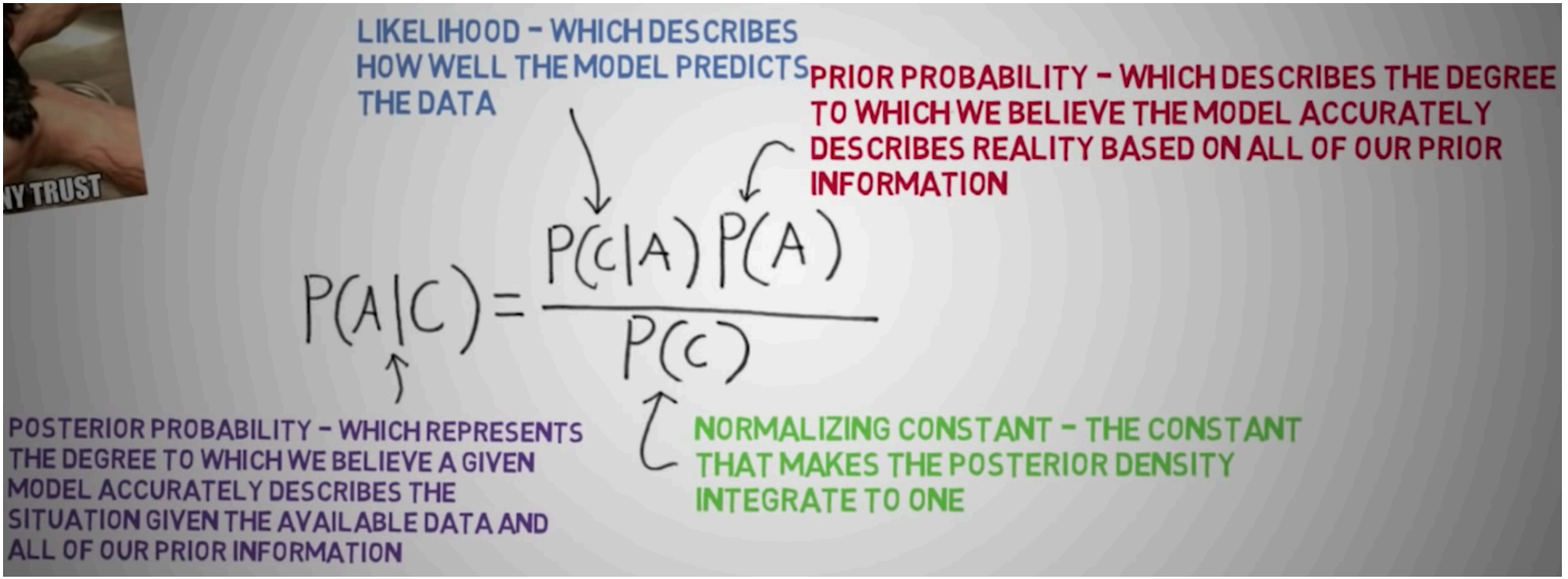
Probability we cannot play a game:
> P(Outlook=Sunny | Play=No) = 3/5
> P(Temperature=Cool | Play=No) = 1/5
> P(Humidity=High | Play=No) = 4/5
> P(Wind=Strong | Play=No) = 3/5
> P(Play=No) = 5/14

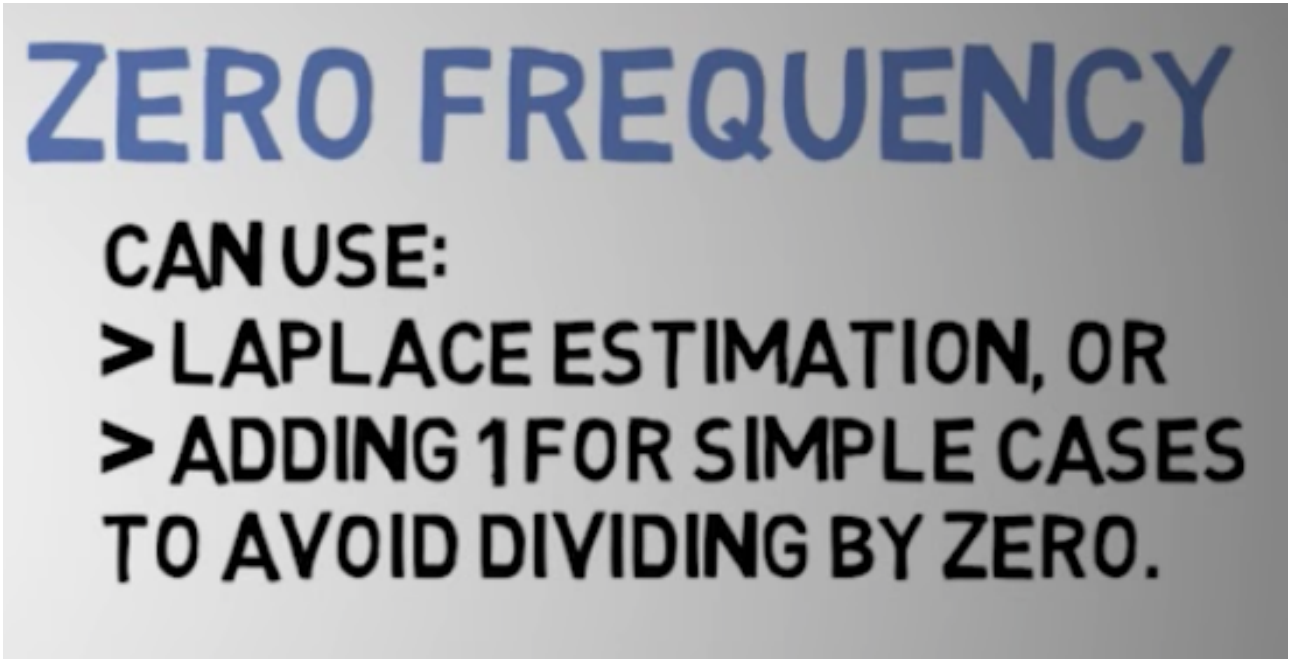
P(X|C)P(C), or P(X|Play=Yes)P(Play=Yes)

P(X | Play=Yes)P(Play=Yes) = (2/9) * (3/9) * (3/9) * (3/9) * (9/14) = 0.0053

P(X|Play=No)P(Play=No) = (3/5) * (1/5) * (4/5) * (3/5) * (5/14) = 0.0206

- These numbers are not probabilities. They are likelihood values weighted by the prior probabilities.
- You can by the evidence to normalize.


Y TRUST
LIKELIHOOD - WHICH DESCRIBES HOW WELL THE MODEL PREDICTS THE DATA
PRIOR PROBABILITY - WHICH DESCRIBES THE DEGREE TO WHICH WE BELIEVE THE MODEL ACCURATELY DESCRIBES REALITY BASED ON ALL OF OUR PRIOR INFORMATION
P(A|C) = P(C|A) P(A) / P(C)
POSTERIOR PROBABILITY - WHICH REPRESENTS THE DEGREE TO WHICH WE BELIEVE A GIVEN MODEL ACCURATELY DESCRIBES THE SITUATION GIVEN THE AVAILABLE DATA AND ALL OF OUR PRIOR INFORMATION
NORMALIZING CONSTANT - THE CONSTANT THAT MAKES THE POSTERIOR DENSITY INTEGRATE TO ONE



If the frequency of a feature is zero for a given class, you can add one to all numbers in all the frequency tables.

What if features are continuous?

e.g., character recognition: X_i is intensity at i^{th} pixel



Gaussian Naïve Bayes (GNB):

$$P(X_i = x \mid Y = y_k) = \frac{1}{\sigma_{ik}\sqrt{2\pi}} \; e^{\frac{-(x-\mu_{ik})^2}{2\sigma_{ik}^2}}$$

Different mean and variance for each class k and each pixel i.

Sometimes assume variance

- is independent of Y (i.e., σ_i),
- or independent of X_i (i.e., σ_k)
- or both (i.e., σ)

Estimating parameters: Y discrete, X_i continuous

$$h_{NB}(\mathbf{x}) = \arg\max_y P(y) \prod_i P(X_i = x_i | y)$$

$$\approx \arg\max_k \hat{P}(Y = k) \prod_i \mathcal{N}(\hat{\mu}_{ik}, \hat{\sigma}_{ik})$$

$$\hat{\mu}_{MLE} = \frac{1}{N} \sum_{j=1}^{N} x_j$$

$$\hat{\sigma}^2_{unbiased} = \frac{1}{N-1} \sum_{j=1}^{N} (x_j - \hat{\mu})^2$$

Estimating parameters: Y discrete, X_i continuous

Maximum likelihood estimates:

$$\hat{\mu}_{MLE} = \frac{1}{N}\sum_{j=1}^{N} x_j$$

$$\hat{\mu}_{ik} = \frac{1}{\sum_j \delta(Y^j = y_k)} \sum_j X_i^j \delta(Y^j = y_k)$$

k^{th} class

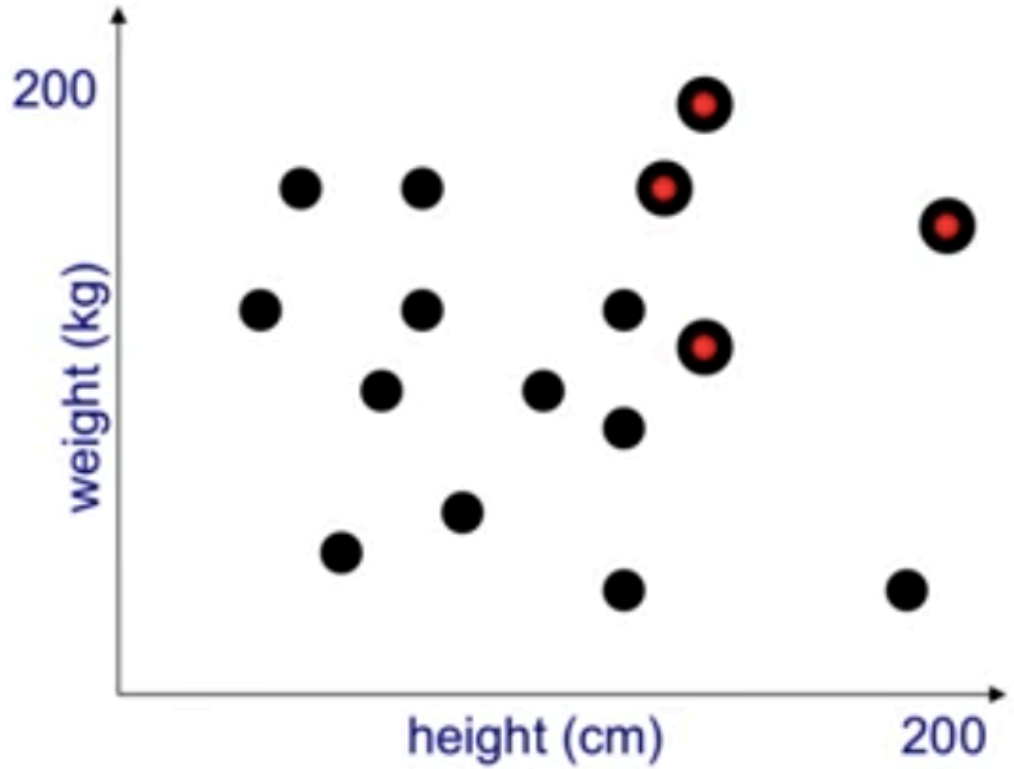
j^{th} training image

i^{th} pixel in j^{th} training image

$$\hat{\sigma}^2_{unbiased} = \frac{1}{N-1}\sum_{j=1}^{N} (x_j - \hat{\mu})^2$$

$$\hat{\sigma}^2_{ik} = \frac{1}{\sum_j \delta(Y^j = y_k) - 1} \sum_j (X_i^j - \hat{\mu}_{ik})^2 \delta(Y^j = y_k)$$

Continuous example

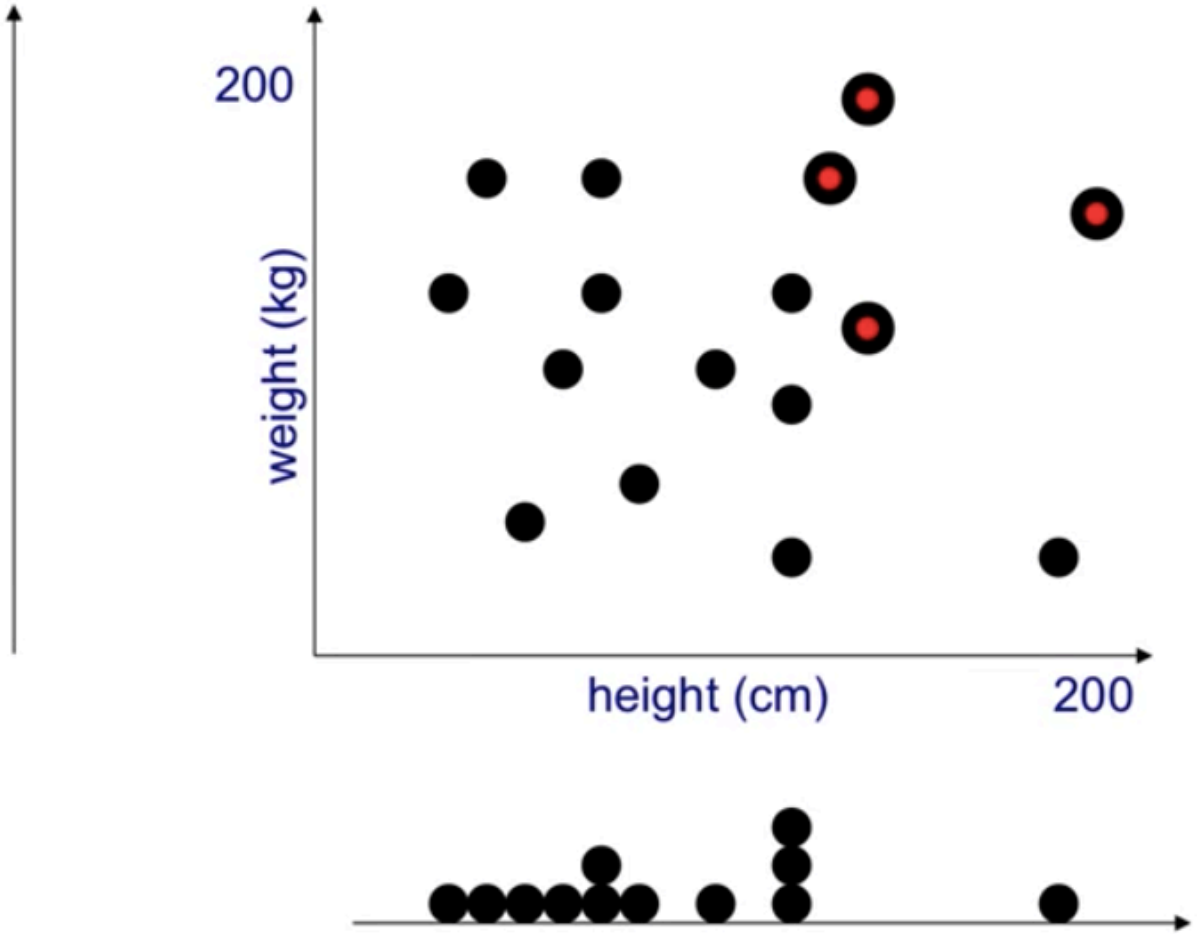


Two classes:

Adult (red)
Child (black)

Continuous example

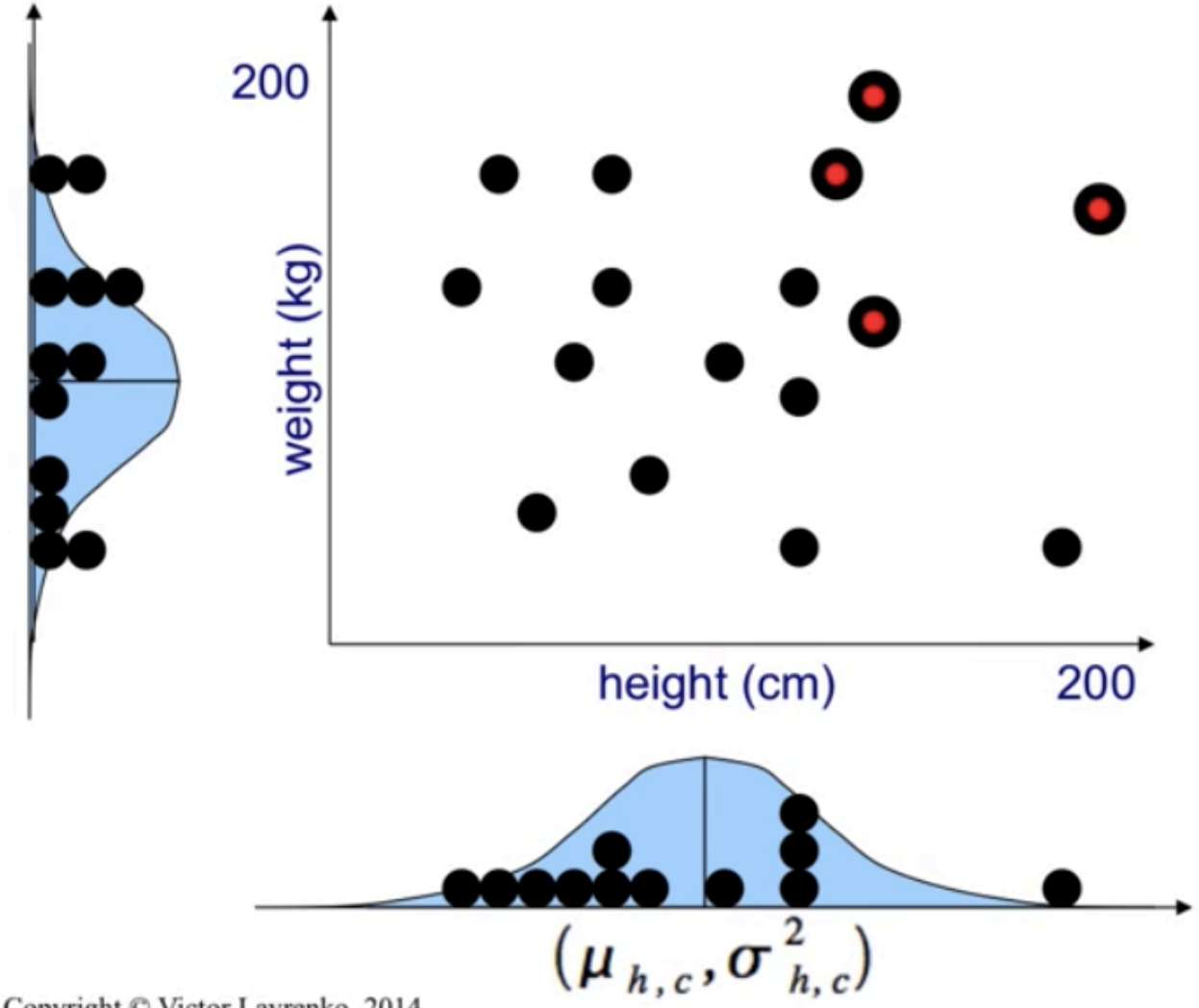
$$P(a)=\frac{4}{4+12}=0.25\ ;\ P(c)=0.75$$



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Continuous example

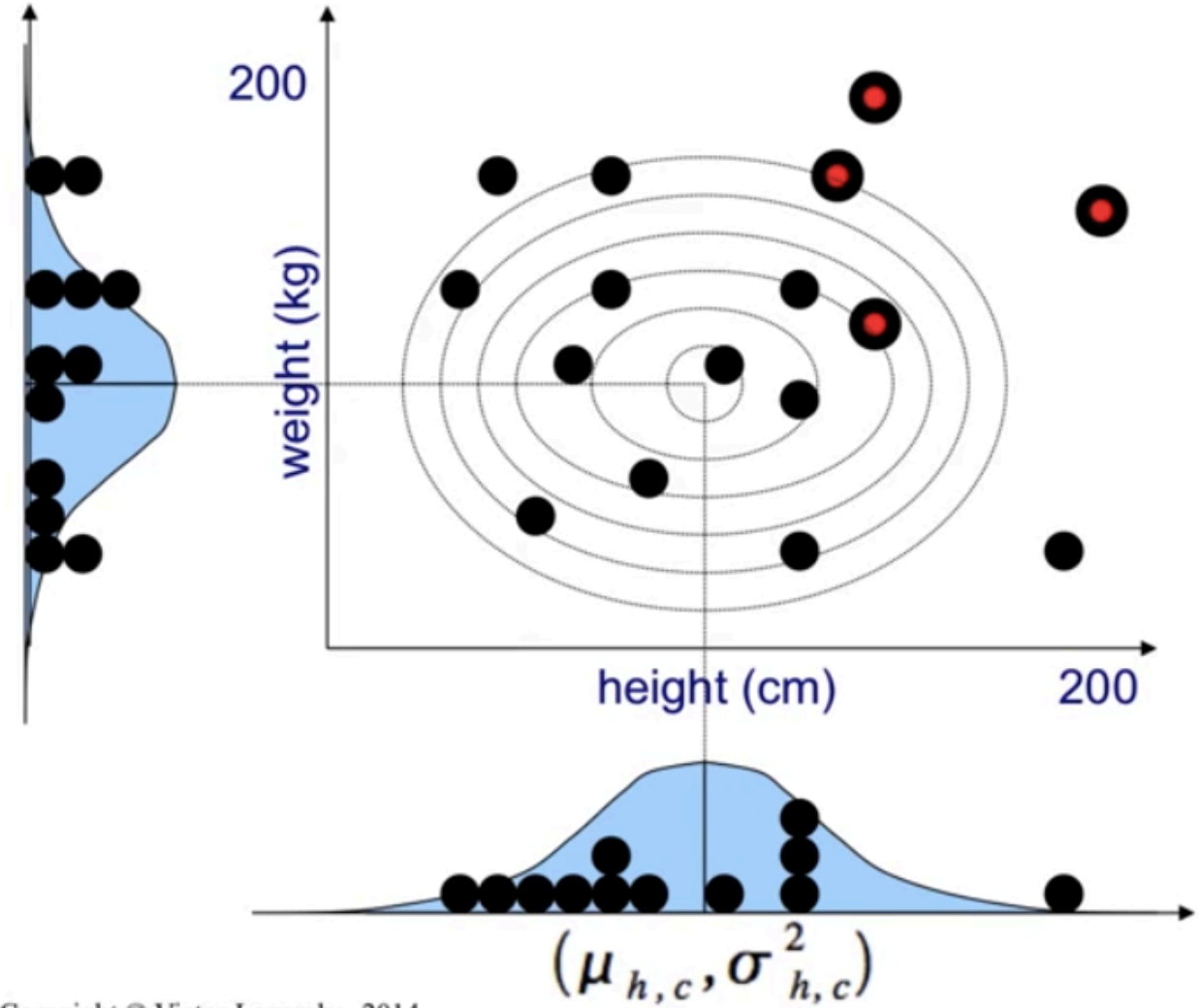
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Continuous example

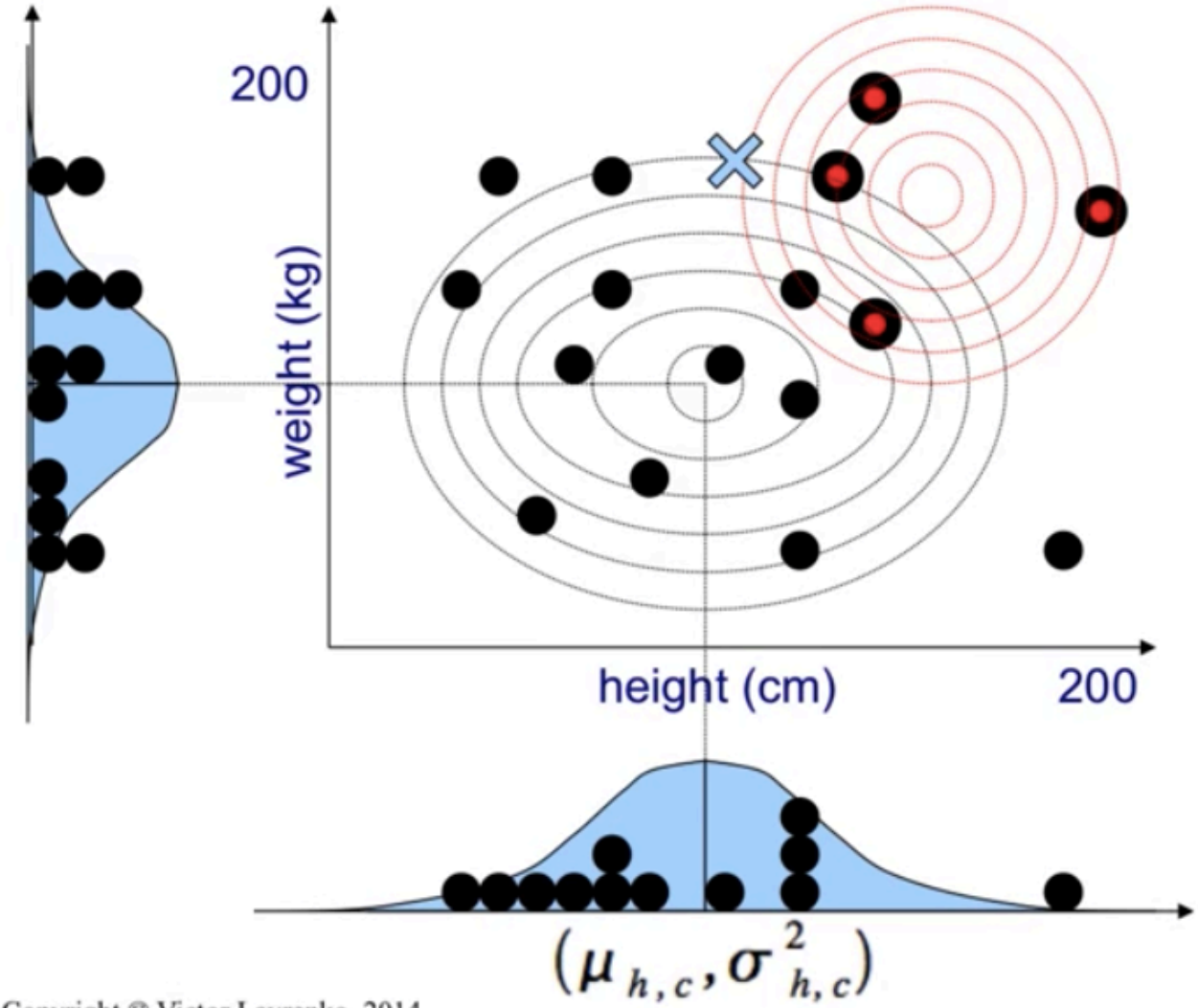
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Continuous example

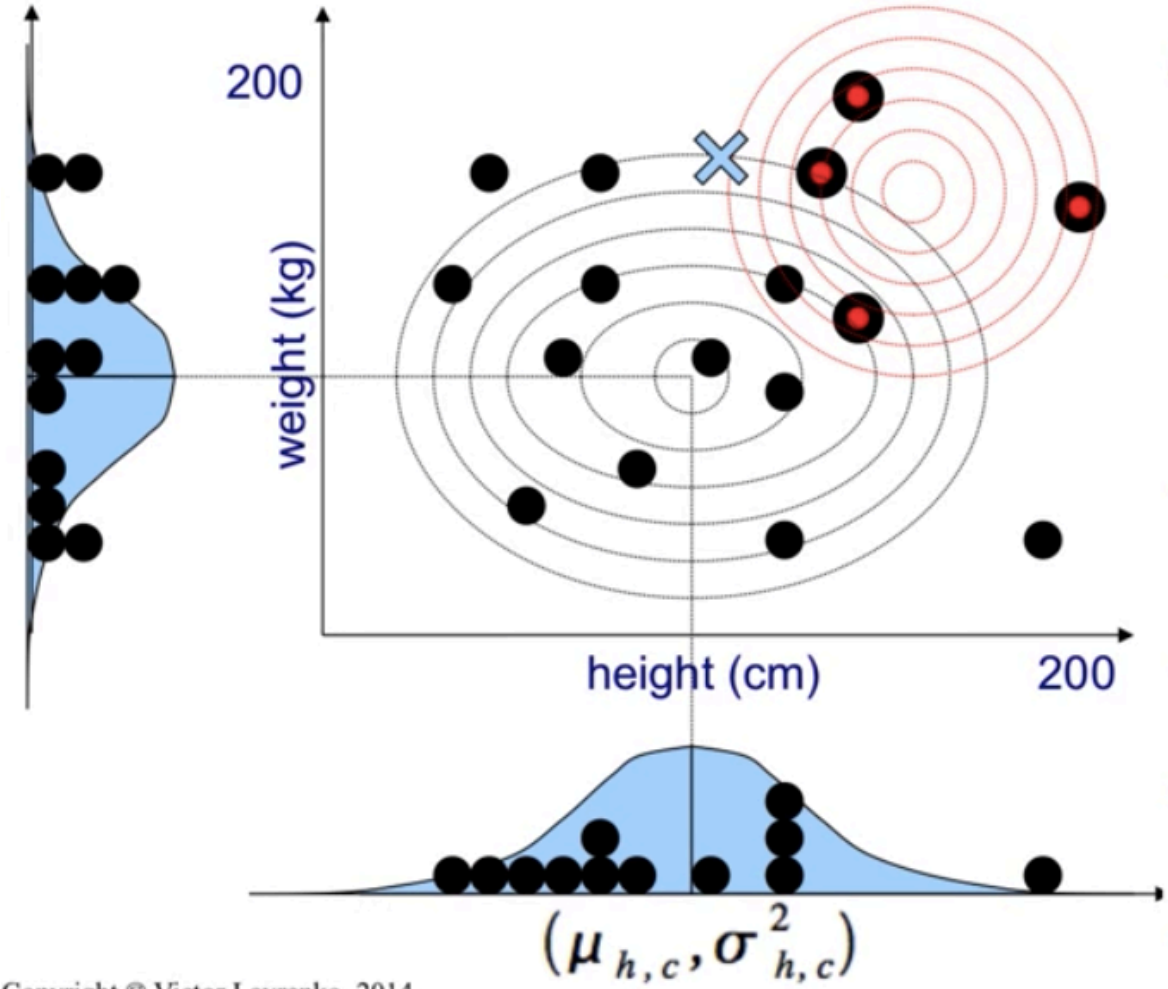
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Continuous example

$$P(a)=\frac{4}{4+12}=0.25\ ;\ P(c)=0.75$$



$$p(h_x|c)=\frac{1}{\sqrt{2\pi\,\sigma^2_{h,c}}}\exp-\frac{1}{2}\left(\frac{(h_x-\mu_{h,c})^2}{\sigma^2_{h,c}}\right)$$

$$p(w_x|c)=\frac{1}{\sqrt{2\pi\,\sigma^2_{w,c}}}\exp-\frac{1}{2}\left(\frac{(w_x-\mu_{w,c})^2}{\sigma^2_{w,c}}\right)$$

$$p(h_x|a)=\frac{1}{\sqrt{2\pi\,\sigma^2_{h,a}}}\exp-\frac{1}{2}\left(\frac{(h_x-\mu_{h,a})^2}{\sigma^2_{h,a}}\right)$$

$$p(w_x|a)=\frac{1}{\sqrt{2\pi\,\sigma^2_{w,a}}}\exp-\frac{1}{2}\left(\frac{(w_x-\mu_{w,a})^2}{\sigma^2_{w,a}}\right)$$

$$P(x|a)=p(h_x|a)\,p(w_x|a)$$

$$P(x|c)=p(h_x|c)\,p(w_x|c)$$

$$P(a|x)=\frac{P(x|a)P(a)}{P(x|a)P(a)+P(x|c)P(c)}$$

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Example: Spam Filter

- Input: an email
- Output: spam/ham

- Setup:
  - Get a large collection of example emails, each labeled "spam" or "ham"
  - Note: someone has to hand label all this data!
  - Want to learn to predict labels of new, future emails

- Features: The attributes used to make the ham / spam decision
  - Words: FREE!
  - Text Patterns: $dd, CAPS
  - Non-text: SenderInContacts
  - …

Dear Sir.

First, I must solicit your confidence in this transaction, this is by virture of its nature as being utterly confidencial and top secret. …

TO BE REMOVED FROM FUTURE MAILINGS, SIMPLY REPLY TO THIS MESSAGE AND PUT "REMOVE" IN THE SUBJECT.

99 MILLION EMAIL ADDRESSES FOR ONLY $99

Ok, Iknow this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.

Example

Email classification

Suppose our vocabulary contains three words a, b and c, and we use a multivariate Bernoulli model for our e-mails, with parameters

$$\boldsymbol{\theta}^{\oplus} = (0.5, 0.67, 0.33) \qquad \boldsymbol{\theta}^{\ominus} = (0.67, 0.33, 0.33)$$

This means, for example, that the presence of b is twice as likely in spam ($+$), compared with ham.

The e-mail to be classified contains words a and b but not c, and hence is described by the bit vector $\mathbf{x} = (1, 1, 0)$. We obtain likelihoods

$$P(\mathbf{x}|\oplus) = 0.5 \cdot 0.67 \cdot (1 - 0.33) = 0.222 \qquad P(\mathbf{x}|\ominus) = 0.67 \cdot 0.33 \cdot (1 - 0.33) = 0.148$$

The ML classification of $\mathbf{x}$ is thus spam.

Example

Email classification: training data

E-mail	a?	b?	c?	Class
e_1	0	1	0	+
e_2	0	1	1	+
e_3	1	0	0	+
e_4	1	1	0	+
e_5	1	1	0	−
e_6	1	0	1	−
e_7	1	0	0	−
e_8	0	0	0	−

What are the parameters of the model?

Example

Email classification: training data

E-mail	*a*?	*b*?	*c*?	Class
e_1	0	1	0	+
e_2	0	1	1	+
e_3	1	0	0	+
e_4	1	1	0	+
e_5	1	1	0	−
e_6	1	0	1	−
e_7	1	0	0	−
e_8	0	0	0	−

What are the parameters of the model?

P(+) = 0.5, P(-) = 0.5

$$\hat{P}(y) = \frac{|\{i : y_i = y\}|}{n}$$

P(a|+) = 0.5, P(a|-) = 0.75
P(b|+) = 0.75, P(b|-)= 0.25
P(c|+) = 0.25, P(c|-)= 0.25

$$\hat{P}(x_i|y) = \frac{\hat{P}(x_i, y)}{\hat{P}(y)} = \frac{|\{i : X_{ij} = x_i, y_i = y\}|/n}{|\{i : y_i = y\}|/n}$$

Comments on Naïve Bayes

Usually features are not conditionally independent, i.e.

$$P(\mathbf{X}|Y) \neq P(x_1|Y)P(x_2|Y), \ldots, P(x_d|Y)$$

And yet, one of the most widely used classifiers. Easy to train!

It often performs well even when the assumption is violated.

Domingos, P., & Pazzani, M. (1997). Beyond Independence: Conditions for the Optimality of the Simple Bayesian Classifier. *Machine Learning*. 29, 103-130.

When there are few training examples

What if you never see a training example where x_1=a when y=spam?

P(x | spam) = P(a | spam) P(b | spam) P(c | spam) = 0

What to do?

Add "virtual" examples for which x_1=a when y=spam.

Summary

Naïve Bayes classifier:

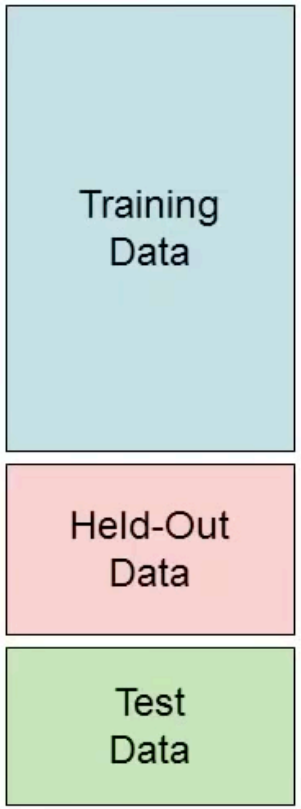
- What's the assumption
- Why we make it
- How we learn it

Naïve Bayes for discrete data

Gaussian naïve Bayes

Important Concepts

- Data: labeled instances, e.g. emails marked spam/ham
  - Training set
  - Held out set
  - Test set
- Features: attribute-value pairs which characterize each x
- Experimentation cycle
  - Learn parameters (e.g. model probabilities) on training set
  - (Tune hyperparameters on held-out set)
  - Compute accuracy of test set
  - Very important: never "peek" at the test set!
- Evaluation
  - Accuracy: fraction of instances predicted correctly
- Overfitting and generalization
  - Want a classifier which does well on *test* data
  - Overfitting: fitting the training data very closely, but not generalizing well
  - We'll investigate overfitting and generalization formally in a few lectures





References and Links

- Text classification Using Naive Bayes
  - https://www.youtube.com/watch?v=EGKeC2S44Rs&t=604s

Learning to Classify Text

Target concept *Interesting*? : *Document* $\rightarrow \{+, -\}$

1. Represent each document by vector of words
   - one attribute per word position in document
2. Learning: Use training examples to estimate
   - $P(+)$
   - $P(-)$
   - $P(doc|+)$
   - $P(doc|-)$

Naive Bayes conditional independence assumption

$$P(doc|v_j) = \prod_{i=1}^{length(doc)} P(a_i = w_k|v_j)$$

where $P(a_i = w_k|v_j)$ is probability that word in position i is w_k, given v_j

one more assumption: $P(a_i = w_k|v_j) = P(a_m = w_k|v_j), \forall i, m$

Naive Bayes for Text Classification
An Example: Movie Reviews

You have a set of reviews (documents) and a clasisfication

Doc	Text	Class
1	I loved the movie	+
2	I hated the movie	-
3	a great movie. good movie	+
4	poor acting	-
5	great acting. a good movie	+

10 Unique words:

<I, loved,the,movie,hated,a,great,poor,acting,good>

Naive Bayes for Text Classification
An Example: Movie Reviews

Convert the documents into feature sets, where the attributes are possible words, and the values are the number of times a word occurs in the given document.

Doc	I	loved	the	movie	hated	a	great	poor	acting	good	Class
1	1	1	1	1							+
2	1		1	1	1						-
3				2		1	1			1	+
4								1	1		-
5				1		1	1		1	1	+

Naive Bayes for Text Classification
An Example: Movie Reviews

Convert the documents into feature sets, where the attributes are possible words, and the values are the number of times a word occurs in the given document.

Doc	I	loved	the	movie	hated	a	great	poor	acting	good	Class
1	1	1	1	1							+
2	1		1	1	1						-
3				2		1	1			1	+
4								1	1		-
5				1		1	1		1	1	+

Let's look at the probabilities per outcome (+ or -)

Naive Bayes for Text Classification
An Example: Movie Reviews

Documents with positive outcomes.

Doc	I	loved	the	movie	hated	a	great	poor	acting	good	Class
1	1	1	1	1							+
3				2		1	1			1	+
5				1		1	1		1	1	+

$p(+) = \frac{3}{5} = 0.6$

Compute:$p(I|+)$; $p(love|+)$; $p(the|+)$; $p(movie|+)$;
$p(a|+)$; $p(great|+)$; $p(acting|+)$; $p(good|+)$

Let n be the number of words in the (+) case: 14 . n_k the number of times word k occurs in these cases (+)

Let $p(w_k|+) = \frac{n_k+1}{n+|Vocabulary|}$

Naive Bayes for Text Classification
An Example: Movie Reviews

Documents with positive outcomes.

Doc	I	loved	the	movie	hated	a	great	poor	acting	good	Class
1	1	1	1	1							+
3				2		1	1			1	+
5				1		1	1		1	1	+

$$p(+) = \frac{3}{5} = 0.6;\ p(w_k|+) = \frac{n_k+1}{n+|Vocabulary|}$$

$$p(I|+) : \frac{1+1}{14+10} = 0.0833;$$
$$p(the|+) = \frac{1+1}{14+10} = 0.0833;$$
$$p(a|+) = \frac{2+1}{14+10} = 0.125;$$
$$p(acting|+) = \frac{1+1}{14+10} = 0.0833;$$
$$p(hated|+) = \frac{0+1}{14+10} = 0.0417$$

$$p(loved|+) = \frac{1+1}{14+10} = 0.0833$$
$$p(movie|+) = \frac{5+1}{14+10} = 0.2083;$$
$$p(great|+) = \frac{2+1}{14+10} = 0.125;$$
$$p(good|+) = \frac{2+1}{14+10} = 0.125$$
$$p(poor|+) = \frac{0+1}{14+10} = 0.0417$$

Naive Bayes for Text Classification
An Example: Movie Reviews

Now, let's look at the negative examples.

Doc	I	loved	the	movie	hated	a	great	poor	acting	good	Class
2	1		1	1	1						-
4								1	1		-

$$p(-) = \frac{2}{5} = 0.4$$

$$p(I|-) : \frac{1+1}{6+10} = 0.125; \qquad p(the|-) = \frac{1+1}{6+10} = 0.125$$
$$p(movie|-) = \frac{1+1}{6+10} = 0.125; \qquad p(hated|-) = \frac{1+1}{6+10} = 0.125;$$
$$p(poor|-) = \frac{1+1}{6+10} = 0.125 \qquad p(acting|-) = \frac{1+1}{6+10} = 0.125;$$
$$p(loved|-) = \frac{0+1}{6+10} = 0.0625 \qquad p(a|-) = \frac{0+1}{6+10} = 0.0625$$
$$p(great|-) = \frac{0+1}{6+10} = 0.0625 \qquad p(good|-) = \frac{0+1}{6+10} = 0.0625$$

Naive Bayes for Text Classification
An Example:Movie Reviews

Now that we've trained our classifier,

Let's classify a new sentence according to:

$$V_{NB} = \underset{v_j \in V}{\operatorname{argmax}} P(v_j) \prod_{w \in words} P(w|v_j)$$

where V stands for "value" or "class"

"I hated the poor acting"

if $v_j = +$; $p(+)p(I|+)p(hated|+)p(the|+)p(poor|+)p(acting|+) = 6.03 \times 10^{-7}$

if $v_j = -$; $p(-)p(I|-)p(hated|-)p(the|-)p(poor|-)p(acting|-) = 1.22 \times 10^{-5}$

Learning to Classify Text

Learn_naive_Bayes_text(*Examples*, *V*)

1. *collect all words and other tokens that occur in Examples*

- *Vocabulary* ← all distinct words and other tokens in *Examples*

2. *calculate the required* $P(v_j)$ *and* $P(w_k|v_j)$ *probability terms*

- For each target value v_j in V do
  - $docs_j \leftarrow$ subset of *Examples* for which the target value is v_j
  - $P(v_j) \leftarrow \frac{|docs_j|}{|Examples|}$
  - $Text_j \leftarrow$ a single document created by concatenating all members of $docs_j$
  - $n \leftarrow$ total number of words in $Text_j$ (counting duplicate words multiple times)
  - for each word w_k in *Vocabulary*
    - $n_k \leftarrow$ number of times word w_k occurs in $Text_j$
    - $P(w_k|v_j) \leftarrow \frac{n_k+1}{n+|Vocabulary|}$

Learning to Classify Text

Classify_naive_Bayes_text(*Doc*)

- *positions* ← all word positions in *Doc* that contain tokens found in *Vocabulary*
- Return v_{NB}, where

$$v_{NB} = \underset{v_j \in V}{\operatorname{argmax}} P(v_j) \prod_{i \in positions} P(a_i | v_j)$$